

HCI — ASSIGNMENT N. 2

Design and prototyping of an application using a human-centered approach

Final Presentation and Demo

Project Title: VeggieWorld

Group: André Butuc, Bruna Simões & Gonçalo Silva

Lab Class: P5





INTRODUCTION

- VeggieWorld is a mobile application to help monitor and manage children's vegetable consumption and raise awareness regarding healthy meals.

OBJECTIVES

- Provide a user-friendly platform to manage children's vegetable consumption
- Contribute to vegetable consumption awareness, through healthy meals sharing

INTERVIEWS

■ Interviewee 1

Adult male, 43 years old,
living in Aveiro. Currently,
working in Accounting.

Married with 2 children.

Feedback:

- His children had a tough time eating vegetables and would sometimes "fake eat" them.
- Because of always getting late from work, he couldn't cook healthy meals and would make fast pre-cooked meals.

■ Interviewee 2

Adult female, 45 years old,
living in Aveiro. Currently,
working as a Mechanical
Engineer.

Married with 3 children.

Feedback:

- She was faced several times with strict diets and would usually lose count of her children's daily progress.
- When growing up, she was always encouraged to cook and eat healthy meals

- **Demographics:**
 - 36 years old
 - Accounting major
 - Married
 - Mother of 2 children, 6 and 8-year-old
- **Goals and tasks:**
 - Track her children's vegetable consumption
 - Get to know different and fun recipes that involve vegetables
 - Set up the child's interactive game side of the app
- **Environment:**
 - She works on a hybrid remote-physical schedule, entering her shift at 10:00 and leaving at 17:00 with a one-hour break for lunch. In terms of her ease with technology, she has only mastered her work computer, specifically Excel, however, she also has a smartphone with fun applications she likes to use. Whenever she cooks dinner, her children are terrible when it comes to eating vegetables.

PERSONA

Persona: Invested and concerned parent

Fictional name: Sarah Smith

Job title: Accountant and full time Mother





SCENARIO 1

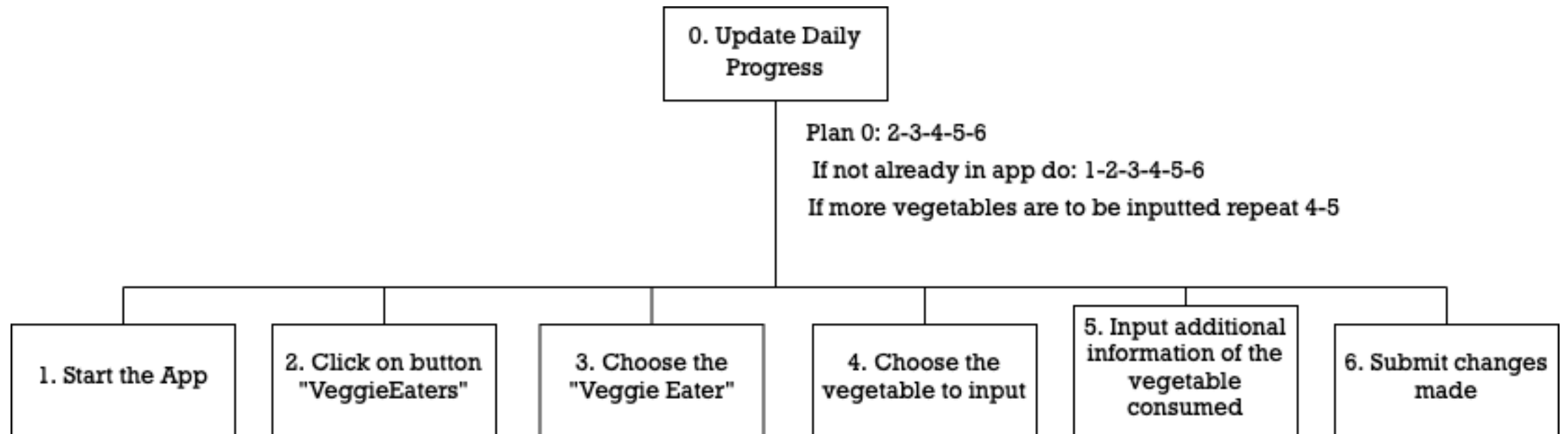
- After dinner, Sarah Smith wants to update her children's daily progress in the application, so she opens the app and goes to the page "VeggieEaters" where she can select one of the children she has registered and update the daily progress.



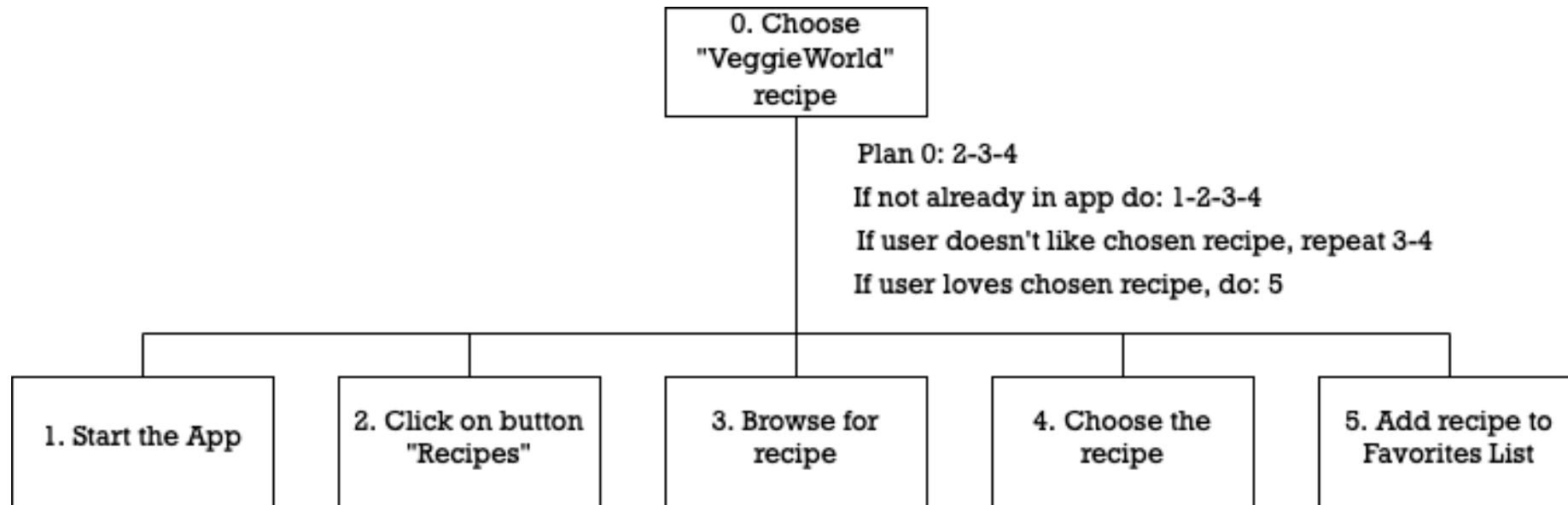
SCENARIO 2

- Scenario 2 – Sarah Smith has lately been scarce on ideas of what healthy and fun meals to cook for her children, knowing that her time is limited she's struggling to find reliable and quick recipes to cook for dinner. Therefore, Sarah opens the VeggieWorld app and goes to the “Recipes” page where she can scan the type of dish she would want to cook and choose a dish recipe.

TASK 1 – UPDATE DAILY PROGRESS



TASK 2 – CHOOSE VEGGIEWORLD RECIPE



REQUIREMENTS

Non-functional

- Usability:
 - Clean and simplistic interface
 - User can learn using the system quickly and easily
- Performance:
 - Be compatible with android 3.0/iOS 4 and above
 - Run, with restrictions, without internet connection.
- Interface
 - Dark Mode
 - App's Text must be visible from 1 meter away

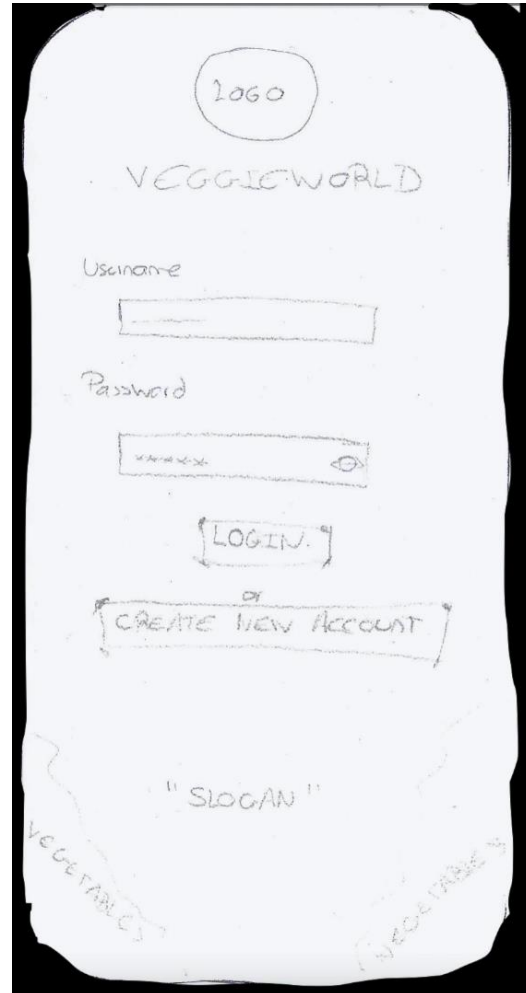
Functional

- Accomplishable:
 - Update VeggieEater's daily progress
 - View VeggieEater's statistics
 - Read and Share recipes
 - Register VeggieEaters
 - Create team of VeggieEaters
 - Define goals and landmarks
 - View and edit VeggieEater's history
- Non-Accomplishable:
 - Define reward system for VeggieEater
 - Generate "VeggieEaters" credentials to the App
 - Web/Browser Version of VeggieWorld

LOW FIDELITY PROTOTYPE

- Paper prototype (number of pages made: 16)
- Core functionalities presented:
 - Update VeggieEater's daily progress
 - Browse Recipe and add it to favourite recipes list
- The paper prototype implemented the:
 - Login page
 - Main page
 - Veggie Eaters page
 - Veggie Eater page
 - Recipes page
 - Profile page
 - Q&A page
 - "Feature Not Implemented" page
 - Confirmation pop-up messages
 - Vegetable modal
 - Recipe Modal

LOW FIDELITY PROTOTYPE



LFP USER EVALUATION

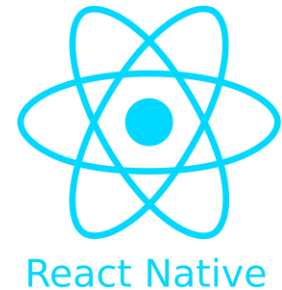
- Goal:
 - Evaluate the design
 - Evaluate application's ease of use
 - Access the logical flow and error prone locations
- Persona: Invested and Concerned Parent
- Scenarios:
 - "Update Veggie Eater's Daily Progress"
 - "Browse and Pick Recipe"
- Number of participants: 8
 - 7 students of IHC P5
 - 1 IHC professor

LFP USER EVALUATION

- Main results:
 - Changed the position of “Add Recipe” button;
 - Removed “plus” and “minus” icons from “Veggie Eaters” page;
 - Added labels to give the user more directions towards the goal;
 - Moved the vegetable search bar closer to the vegetables.

PLATFORM USED FOR THE FUNCTIONAL PROTOTYPE

- React Native:
 - Open-source UI software framework created by Meta Platforms.
 - Good learning curve and easy to use. Broad documentation and GitHub community.
- Android Studio:
 - Used to integrate React Native with its emulator
- Android Phone and Emulators:
 - To test app's performance and feel on a real Android phone



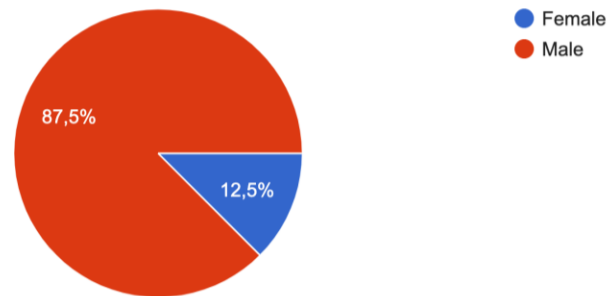
USER EVALUATION: FUNCTIONAL PROTOTYPE

- The main goals of the user evaluation were the same as the ones in the paper prototype evaluation, however, this time there was more focus on the design as well.
- Persona: Invested and Concerned Parent
- Scenario Tasks:
 - Scenario Task 1 – Add Vegetables and Remove them from the Arla's Daily Progress
 - Scenario Task 2 – Add a recipe to your favourites list and then add the favourite recipe to Carlos' daily progress.

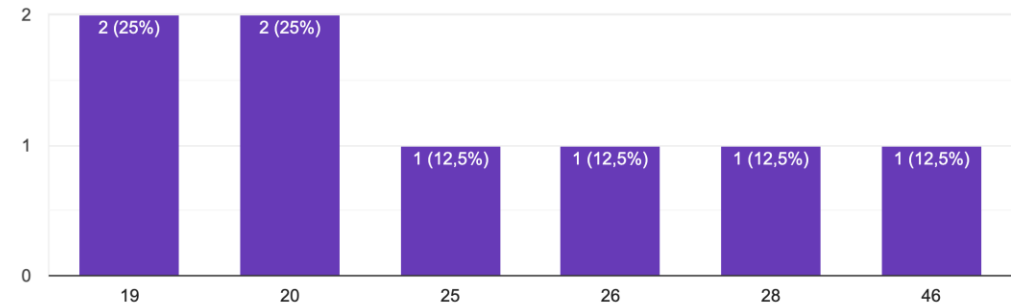
USER EVALUATION: FUNCTIONAL PROTOTYPE

■ Participants:

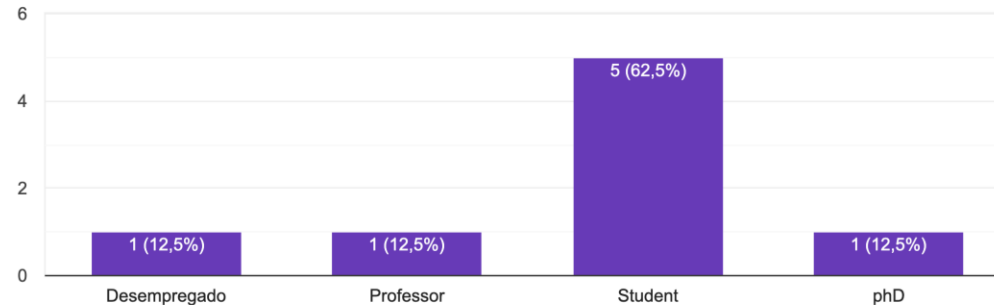
Gender
8 respostas



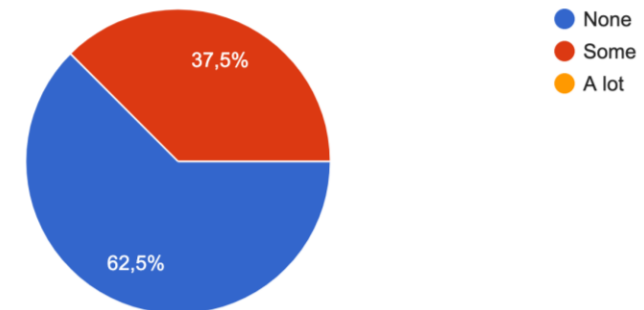
Age
8 respostas



Profession
8 respostas



Previous experience with this type of application/system:
8 respostas

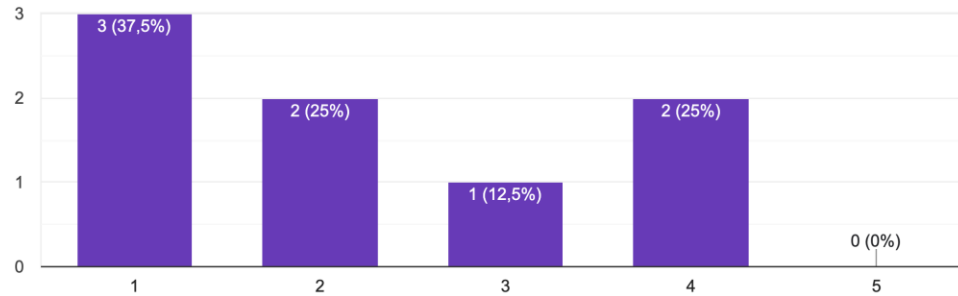


USER EVALUATION: FUNCTIONAL PROTOTYPE

- Feedback (1: Completely Agree; 5: Completely Disagree)

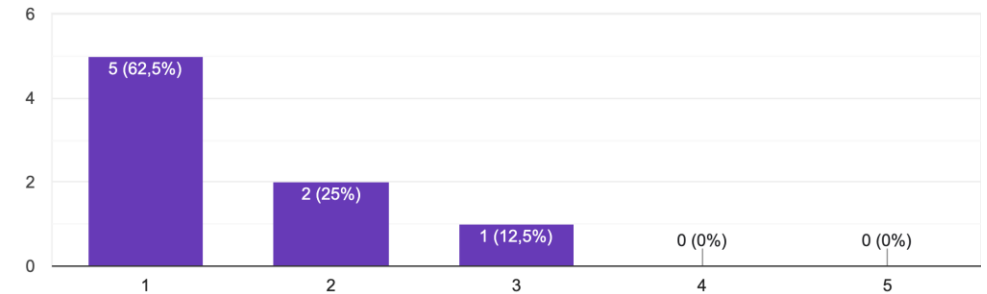
I think that I would like to use this system frequently.

8 respostas



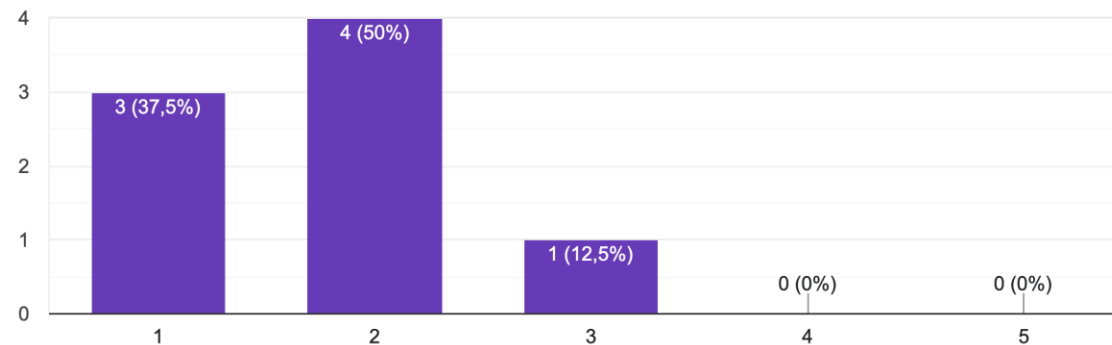
I thought the system was easy to use.

8 respostas



I felt very confident using the system.

8 respostas



USER EVALUATION: FUNCTIONAL PROTOTYPE

- Problems found:
 - Background-image makes app less readable (labels+text);
 - “Remove Vegetable Button Icon” is too small;
 - “Recipe Button” in Recipes Page is misleading;
 - Vegetable State CheckBox;
 - Dynamic Search isn’t implemented;
 - Confirmation Pop-up Message is not visible in the middle of the screen;
 - Needs the existence of the logical flow: Add Vegetables -> Select Veggie Eaters to Update.

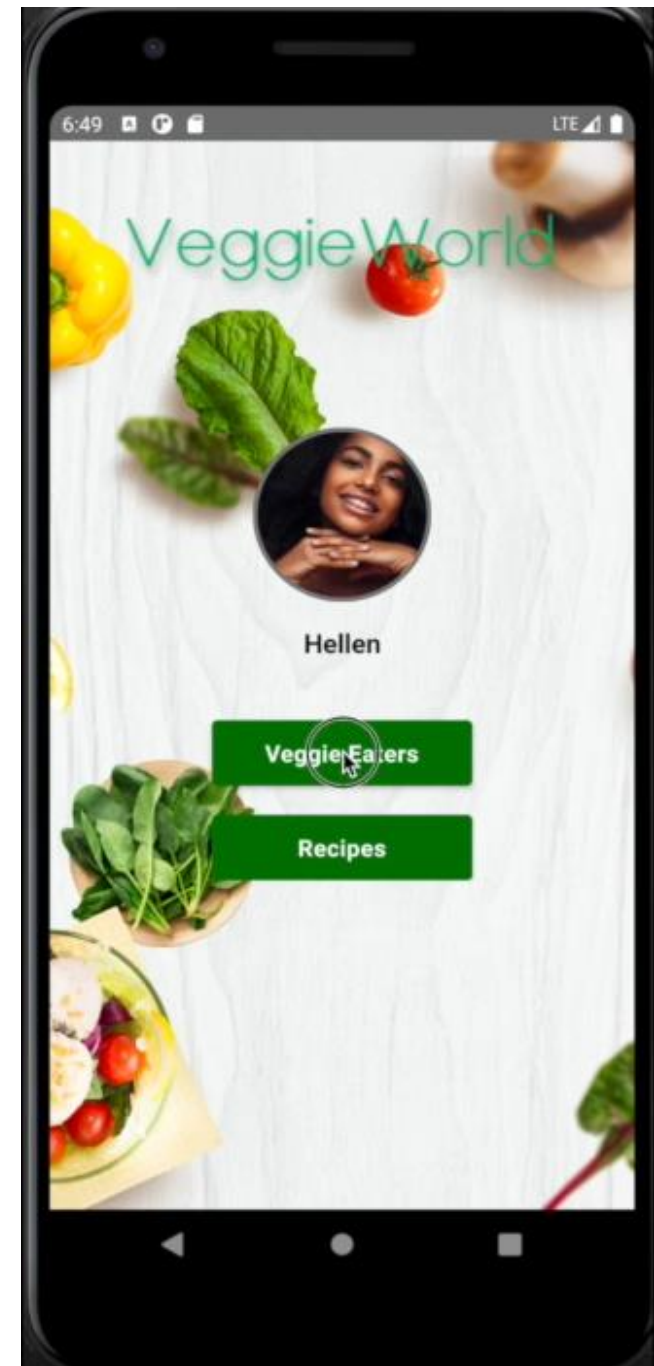
HEURISTIC EVALUATION OF THE FUNCTIONAL PROTOTYPE

Usability problem	Heuristic	Severity Degree ^[1]
It's difficult to read labels	#6 Recognition rather than recall	4
No Dynamic Search	#2 Match between system and the real world	3
Unfriendly Logical Flow on Daily Progress Update	#7 Flexibility and efficiency of use	2
Can't dynamically change recipes favourites list	#2 Match between system and the real world	3
Can't add Vegetables from Recipes Page	#7 Flexibility and efficiency of use	2

[1] - Median

DEMO

- **Tasks:**
 - Add Vegetable to Veggie Eater's Daily Progress List;
 - Remove recently added Vegetable;
 - Browse and pick Favorite Recipe
 - Add recipe to Veggie Eater's Daily Progress



FUTURE WORK

- To be implemented:
 - Logical flow: Select Vegetables → Select Veggie Eaters to Update
 - Statistics Page
 - Profile Page
 - Login Page
 - Edit Veggie Eater's History
 - Add, remove and edit Veggie Eater

ACKNOWLEDGES

- Special thanks to:
 - Participants of the Usability Tests
 - Friends and family
 - IHC's professors

EFFORT OF EACH GROUP MEMBER

- André Butuc – 50%
- Bruna Simões – 25%
- Gonçalo Silva – 25%

QUESTIONS?

